

Machine Learning Mid-Term Project: Single-Task and Multi-Task Learning with "A Century of Portraits" Dataset

Objective

In this mid-term project, you will design and implement two models using PyTorch to analyze the "A Century of Portraits" dataset: (1) a single-task regression model to predict the year a portrait was taken, and (2) a multi-task learning (MTL) neural network to jointly predict the year (regression) and a second task (gender classification or state classification). This project aims to deepen your understanding of classification, regression, multi-task learning, custom dataset creation, and neural network architecture design.

Project Requirements

1. Dataset

You will use the "**A Century of Portraits**" dataset, which contains 37,921 frontal-facing American high school yearbook photos (1905–2013) with accompanying metadata (e.g., year, gender, state, high school district). You are required to:

- **Create a Custom PyTorch Dataset:** Implement a custom `Dataset` class in PyTorch to handle loading and preprocessing of portrait images and metadata for both models. Ensure the dataset supports year prediction (regression) and the chosen second task (gender or state classification).
 - Preprocess images (e.g., resizing to 224×224) and metadata.
 - Handle missing or incomplete data appropriately, if there is any.
 - Using the provided training and test split.

2. Task Definition

You will train two models:

1. **Single-Task Model:** Predict the **year** the portrait was taken as a regression task (output a continuous year value, e.g., 1965.7).
2. **Multi-Task Model:** Jointly predict:
 - **Year** (regression, as above).
 - A second task of your choice:
 - **Gender classification** (binary: male/female), or
 - **State classification** (multi-class: predict the U.S. state where the portrait was taken, ~27 classes based on available metadata).Justify your choice of the second task in your report.

3. Model Architecture

Design two neural network models using PyTorch:

- **Single-Task Model:** A convolutional neural network (CNN) for year regression (e.g., ResNet-18/50 or a custom architecture) with a single output neuron (no activation, for continuous year prediction).
- **Multi-Task Model:** An MTL neural network with a shared backbone (e.g., ResNet convolutional layers) and task-specific heads:

- Year regression head: Single neuron for continuous output.
- Gender/state head: Softmax for classification (2 outputs for gender, ~27 for state).
- **Fine-Tuning:** You are encouraged to use pre-trained models (e.g., ResNet from `torchvision.models`) and fine-tune them. Document your fine-tuning strategy (e.g., partial vs. full fine-tuning).

4. Implementation

Implement training pipelines for both models in PyTorch, including:

- **Data Loading:** Use your custom dataset to load images and metadata.
- **Loss Functions:**
 - Single-Task: L1 (MAE) or MSE loss for year regression.
 - Multi-Task: Combined loss (e.g., weighted sum: $0.7 \times \text{year loss} + 0.3 \times \text{gender/state loss}$). Use cross-entropy for classification tasks.
- **Optimization:** Use an optimizer (e.g., Adam, SGD) with learning rate scheduling.
- **Training/Validation:** Implement loops for both models, monitoring performance on the validation set.
- **Evaluation:** Evaluate both models on the test set using:
 - Year: MAE (mean absolute error in years).
 - Gender: Accuracy, F1 score.
 - State: Accuracy, F1 score.
- Compare single-task vs. multi-task performance, analyzing MTL's impact (e.g., does gender/state improve year prediction?).

5. Deliverables

Submit the following:

- **Code:** A well-documented Jupyter notebook or Python script containing:
 - Custom dataset implementation.
 - Single-task and multi-task model architectures.
 - Training pipelines and evaluation for both models.
- **Report:** A PDF report (2–4 pages) including:
 - **Introduction:** Overview of tasks and their relevance. Also, justify your choice of the second task.
 - **Dataset:** Description of your custom dataset and preprocessing.
 - **Models:** Explanation of architectures, fine-tuning, and MTL design (e.g., loss weighting).
 - **Experiments:** Training details, hyperparameters, and results (include plots/tables: MAE for year; classification metrics for gender/state; training history, etc).
 - **Comparison:** Analyze single-task vs. multi-task performance (e.g., MAE improvement).
 - **Discussion:** Challenges, insights, and potential improvements.

6. Evaluation Criteria

Your project will be evaluated based on:

- **Correctness (35%):** Proper implementation of custom dataset (10%), single-task (10%), and multi-task models (15%).
- **Creativity (5%):** Thoughtful task selection (gender vs. state) and architecture design.
- **Performance (30%):**
 - Undergraduates: MAE < 28 years of year; >85% accuracy for gender; > 10% accuracy for state.

- Graduates: MAE < 22 years of year; >89% accuracy for gender; > 12% accuracy for state.
- **Comparison (5%)**: Clear analysis of single-task vs. multi-task models.
- **Documentation (10%)**: Clarity and completeness of code comments and report.
- **Reproducibility (15%)**: Code runs without errors and produces reported results.

7. Resources

- **Dataset**: The dataset with train/test split is available at TAMU HPRC Launch Cluster under `/scratch/group/p.cis230408.000`
 - More information about the dataset can be found from the "A Century of Portraits" website via <http://shiry.ttic.edu/projects/yearbooks/yearbooks.html>.
- **PyTorch Documentation**: Refer to pytorch.org for dataset creation, model building, and training.
- **Pre-trained Models**: Available via `torchvision.models` for fine-tuning.

8. Submission

- Submit your code and report via Blackboard.
- No late submissions may be accepted.

9. Tips

- **Start Early**: Allow time for data preprocessing, model training, and MTL experimentation.
 - **Time Estimates (total: 9.5-26 hours)**
 - Custom PyTorch Dataset Implementation (1-4 hours)
 - Model Implementation (2-8 hours)
 - Training and Evaluation (6-12 hours)
 - Training both models (~10-50 epochs each): 2-5 hours
 - Debugging training issues: 2-5 hours
 - Evaluation model: 1-2 hours
 - Visualizing result: 1-2 hours
 - Report Writing (.5-2 hours)
- **Fine-Tuning**: Adapt pre-trained models. Experiment with partial vs. full fine-tuning.
- **MTL**: Tune loss weights (e.g., grid search [0.1, 0.9] for auxiliary task). Visualize loss curves and MAE trends.
- **Expected Performance**: In the real-world, we would expect the following performance for this task:
 - **Year Prediction**: Normalize years (e.g., [1905, 2013] to [0, 1]) for regression stability. Expect MAE ~4-7 years (unnormalized). MTL may reduce MAE by ~0.5-1 year (unnormalized).
 - **Gender vs. State**: Gender is easier (~95-98% accuracy) and synergizes better with year prediction. State (~10-25% accuracy) is harder due to noisy metadata.

Good luck, and enjoy exploring regression and multi-task learning with the "A Century of Portraits" dataset!